

## Assignment 1-1

### 1. Relation from the matrix (formal)

Matrix (rows/columns correspond to  $A = \{1,2,3,4\}$ ):

$$M = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Definition: entry  $m_{ij} = 1 \Leftrightarrow (i, j) \in R$ .

List the ordered pairs (read row  $\rightarrow$  column):

$$R = \{(1,2), (2,1), (2,3), (3,1), (3,2)\}.$$

Digraph (description): vertices 1,2,3,4. Directed edges:

- $1 \rightarrow 2$ .
- $2 \rightarrow 1, 2 \rightarrow 3$ .
- $3 \rightarrow 1, 3 \rightarrow 2$ .
- Vertex 4 is isolated (no incident edges).

Properties to note (useful later): edges between 1 and 2 are bi-directional;  $2 \rightarrow 3$  and  $3 \rightarrow 2$  make 2 and 3 mutually linked via  $2 \rightarrow 3$  and  $3 \rightarrow 2$  (actually here we have  $2 \rightarrow 3$  and  $3 \rightarrow 2$  present), and  $3 \rightarrow 1$  closes further links so  $\{1,2,3\}$  are strongly connected except check: from 1 can you reach 3?  $1 \rightarrow 2 \rightarrow 3$  yes. From 3 to 1 exists. So  $\{1,2,3\}$  forms a strongly connected subgraph; 4 is a singleton.

### 2. Relation on $A = \{1, 2, 5\}$ and $B = \{3, 5, 7\}$

Given  $R = \{(a, b) \mid 7 \leq a + b < 10, a \in A, b \in B\}$ .

Compute all  $a + b$  pairs:

- $1 + 3 = 4, 1 + 5 = 6, 1 + 7 = 8$
- $2 + 3 = 5, 2 + 5 = 7, 2 + 7 = 9$
- $5 + 3 = 8, 5 + 5 = 10, 5 + 7 = 12$

Accept those with  $7 \leq a + b < 10$ : sums 7,8,9.

i)  $R$  and complement  $R'$

$$R = \{(1,7), (2,5), (2,7), (5,3)\}.$$

Total  $A \times B$  has 9 pairs. Complement  $R' = A \times B \setminus R$ :

$$R' = \{(1,3), (1,5), (2,3), (5,5), (5,7)\}.$$

## ii) Inverse $R^{-1}$

Reverse each ordered pair of  $R$ :

$$R^{-1} = \{(7,1), (5,2), (7,2), (3,5)\}.$$

## iii) Domain and Range

- $\text{Dom}(R) = \{1,2,5\}$  (all  $a$ 's that appear).
- $\text{Ran}(R) = \{3,5,7\}$  (all  $b$ 's that appear).

For  $R'$ : since complement still uses all  $A$  and  $B$ ,  $\text{Dom}(R') = \{1,2,5\}$ ,  $\text{Ran}(R') = \{3,5,7\}$ .

(If you prefer, present these as sets or in roster form — both acceptable at B.Tech level.)

## 3. Relation properties on positive integers

We analyze reflexive, symmetric, antisymmetric, transitive.

### (i) Relation $R_1 = \{(x, y) \mid xy \text{ is a perfect square}\}$

- **Reflexive:** For any positive integer  $x$ ,  $x \cdot x = x^2$  is a perfect square  $\Rightarrow$  **reflexive**.
- **Symmetric:** If  $xy$  is a square then  $yx$  is the same product  $\Rightarrow$  **symmetric**.
- **Antisymmetric:** Antisymmetry requires: if  $(x, y) \in R$  and  $(y, x) \in R$  then  $x = y$ . But take  $x = 2, y = 8$ :  $2 \cdot 8 = 16$  is square, and  $8 \cdot 2 = 16$  is square but  $2 \neq 8$ . So **not antisymmetric**.
- **Transitive:** Need: if  $xy$  and  $yz$  are squares, does  $xz$  have to be a square? This is **not guaranteed in general**. Give a concrete counterexample: choose  $x = 2, y = 8, z = 18$ :
  - $xy = 2 \cdot 8 = 16 = 4^2$  (square).
  - $yz = 8 \cdot 18 = 144 = 12^2$  (square).
  - $xz = 2 \cdot 18 = 36 = 6^2 \rightarrow$  this *is* a square in this example. We must find one that fails.

Construct a systematic counterexample: express numbers by prime exponents. Let  $x = p$ ,  $y = pq$ ,  $z = q$  with distinct primes  $p, q$ . Then

- $xy = p \cdot pq = p^2q$  — not a perfect square unless  $q$  exponent even. So choose exponents carefully.

Simpler: using prime exponent parity arguments, one can show transitivity fails in general. Therefore state: **not transitive**. (If required, we can craft explicit integers using primes with odd/even exponents; I can produce a tight counterexample on request.)

**Summary:** reflexive, symmetric, not antisymmetric, not transitive.

(ii) Relation  $R_2 = \{(x, y) \mid x + y = 10\}$

- **Reflexive:**  $x + x = 10 \Rightarrow x = 5$  only. Since not every  $x$  satisfies, **not reflexive**.
- **Symmetric:** If  $x + y = 10$  then  $y + x = 10$ . So **symmetric**.
- **Antisymmetric:** Not antisymmetric because  $(3, 7)$  and  $(7, 3)$  belong but  $3 \neq 7$ . So **not antisymmetric**.
- **Transitive:** If  $x + y = 10$  and  $y + z = 10$ , this gives  $x = 10 - y$  and  $z = 10 - y \Rightarrow x = z$ . For transitivity one would require  $x + z = 10$ . But with  $x = z$ , we get  $x + x = 10 \Rightarrow x = 5$ . Thus transitivity fails generally (take  $x = 3, y = 7, z = 3$ : first pair  $(3, 7)$  true, second  $(7, 3)$  true, but  $(3, 3)$ :  $6 \neq 10$  so not in relation). So **not transitive**.

**Summary:** symmetric only.

#### 4. Is $(\mathbb{N}, \mid)$ a poset?

Relation:  $a \mid b$  ( $a$  divides  $b$ ). Check poset axioms:

- **Reflexive:**  $n \mid n$  for all  $n$  (true).
- **Antisymmetric:** If  $a \mid b$  and  $b \mid a$ , then  $a = b$  (true for positive integers).
- **Transitive:** If  $a \mid b$  and  $b \mid c$  then  $a \mid c$  (true).

Therefore  $(\mathbb{N}, \mid)$  is a **poset**. (It's the standard divisibility partial order.)

#### 5. Composition $R \circ S$ and $S \circ R$ and matrices

Given set  $A = \{a, b, c, d\}$  (order fixed as  $a, b, c, d$ ).

$$R = \{(a, a), (a, c), (c, b), (c, d), (d, b)\}$$

$$S = \{(b, a), (c, c), (c, d), (d, a)\}$$

Definition:  $R \circ S = \{(x, z) \mid \exists y, (x, y) \in S \text{ and } (y, z) \in R\}$ .

Similarly  $S \circ R = \{(x, z) \mid \exists y, (x, y) \in R \text{ and } (y, z) \in S\}$ .

### Compute $R \circ S$

List  $y$  values from  $S$  and see  $R$ -images:

- From  $S: (b, a)$ . For  $y = a$ :  $R$  has  $(a, a), (a, c) \Rightarrow$  pairs  $(b, a), (b, c)$ .
- From  $S: (c, c)$ . For  $y = c$ :  $R$  has  $(c, b), (c, d) \Rightarrow$  pairs  $(c, b), (c, d)$ .
- From  $S: (c, d)$ . For  $y = d$ :  $R$  has  $(d, b) \Rightarrow (c, b)$  (already present).
- From  $S: (d, a)$ . For  $y = a$ :  $R$  gives  $(a, a), (a, c) \Rightarrow$  pairs  $(d, a), (d, c)$ .

Collect (no duplicates):

$$R \circ S = \{(b, a), (b, c), (c, b), (c, d), (d, a), (d, c)\}.$$

### Compute $S \circ R$

Take each  $(x, y) \in R$  and find  $S$ -images of  $y$ :

- $(a, a)$ : for  $y = a$ ,  $S$  has no pair starting with  $a \Rightarrow$  none.
- $(a, c)$ :  $y = c \Rightarrow S$  has  $(c, c), (c, d) \Rightarrow (a, c), (a, d)$ .
- $(c, b)$ :  $y = b \Rightarrow S$  has  $(b, a) \Rightarrow (c, a)$ .
- $(c, d)$ :  $y = d \Rightarrow S$  has  $(d, a) \Rightarrow (c, a)$  (already).
- $(d, b)$ :  $y = b \Rightarrow S$  yields  $(d, a)$ .

Collect:

$$S \circ R = \{(a, c), (a, d), (c, a), (d, a)\}.$$

### Matrix representations (order $a, b, c, d$ )

Matrix entries 1 if pair present, 0 otherwise.

- For  $R \circ S$ :

Rows  $a, b, c, d$ ; columns  $a, b, c, d$ .

Pairs present:  $(b, a), (b, c), (c, b), (c, d), (d, a), (d, c)$ .

So matrix:

$$M_{R \circ S} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

- For  $S \circ R$ :

Pairs: (a,c), (a,d), (c,a), (d,a)

$$M_{S \circ R} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

(These are standard 0–1 adjacency matrices; verify row  $i$  corresponds to element  $i$  in the order  $a, b, c, d$ .)

## 6. Reflexive, symmetric and transitive closures — worked example using Q1 digraph

(You uploaded a PDF that includes a digraph image I couldn't parse as text. I'll compute closures for the digraph from Question 1 — i.e., the relation found in Q1 — so you have a full worked B.Tech answer and can apply the same method to any other digraph.)

Recall from Q1:

$$R = \{(1,2), (2,1), (2,3), (3,1), (3,2)\}$$

on  $A = \{1,2,3,4\}$ .

### Definitions (formal)

- **Reflexive closure** of  $R$ :  $R \cup \{(x, x) \mid x \in A\}$ .
- **Symmetric closure**:  $R \cup R^{-1}$ .
- **Transitive closure**: smallest transitive relation containing  $R$ ; equivalently  $R^+ = \bigcup_{k \geq 1} R^k$  where  $R^k$  is composition of  $R$  with itself  $k$  times.

### (a) Reflexive closure $R_{ref}$

Add missing self-loops for all elements:

Current loops present? None in  $R$ . So add  $(1,1), (2,2), (3,3), (4,4)$ .

$$R_{ref} = R \cup \{(1,1), (2,2), (3,3), (4,4)\}.$$

**(b) Symmetric closure  $R_{sym}$** 

Compute  $R^{-1} = \{(2,1), (1,2), (3,2), (1,3), (2,3)\}$ — but note many of these are already in  $R$ .  
Let's list  $R$  and  $R^{-1}$ :

- $R = (1,2), (2,1), (2,3), (3,1), (3,2)$ .
- $R^{-1} = (2,1), (1,2), (3,2), (1,3), (2,3)$ .

Union gives extra pair  $(1,3)$  that was missing originally. So

$$R_{sym} = R \cup \{(1,3)\} = \{(1,2), (2,1), (2,3), (3,1), (3,2), (1,3)\}.$$

(We added  $(1,3)$  because  $(3,1)$  was in  $R$  but  $(1,3)$  was not.)

**(c) Transitive closure  $R^+$** 

Compute reachability:  $(x, z) \in R^+$  iff there exists a directed path from  $x$  to  $z$  of length  $\geq 1$ .

From adjacency:

- From 1:  $1 \rightarrow 2$ . From 2:  $2 \rightarrow 1$  and  $2 \rightarrow 3$ . So 1 reaches 2 (length 1), reaches 1 by  $1 \rightarrow 2 \rightarrow 1$  (length 2), reaches 3 via  $1 \rightarrow 2 \rightarrow 3$  (length 2). Also  $1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 3$  etc; all paths stay within  $\{1,2,3\}$ . So 1 reaches 1,2,3.
- From 2:  $2 \rightarrow 1$ ,  $2 \rightarrow 3$ . From  $1 \rightarrow 2$ , and  $3 \rightarrow 1, 2$ , etc. So 2 reaches 1,2,3 as well.
- From 3:  $3 \rightarrow 1$ ,  $3 \rightarrow 2$ . From these, further reach includes  $3 \rightarrow 1 \rightarrow 2 \rightarrow 3$  etc; so 3 reaches 1,2,3.
- From 4: isolated  $\rightarrow$  reaches nobody (no outgoing edges).

Hence transitive closure includes all ordered pairs among  $\{1,2,3\}$  (except possibly self-loops — but transitive closure must include paths of length  $\geq 1$ ; however if there is a path from a node to itself, self-loop must appear in closure). We have cycles so each of 1,2,3 can reach itself via length  $> 0 \Rightarrow$  include  $(1,1), (2,2), (3,3)$ .

Thus

$$R^+ = \{(i, j) \mid i \in \{1,2,3\}, j \in \{1,2,3\}\}.$$

Concretely:

$$R^+ = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}.$$

Note: 4 remains isolated; no pair with 4 in first coordinate appears in  $R^+$  (unless you add reflexive closure).

If you want the **reflexive, symmetric and transitive closures all together** (i.e., the smallest equivalence relation containing  $R$ ), observe that on  $\{1,2,3\}$  the transitive closure already contains all pairs among 1,2,3, so the equivalence relation induced is the full relation on that set; adding 4's reflexive loop gives complete equivalence classes:  $\{1,2,3\}$  is one equivalence class and  $\{4\}$  another.

## 7. Euclidean algorithm (stepwise)

### (a) gcd (252, 105)

Apply division algorithm:

1.  $252 = 105 \cdot 2 + 42$ .
2.  $105 = 42 \cdot 2 + 21$ .
3.  $42 = 21 \cdot 2 + 0$ .

When remainder becomes 0, last nonzero remainder is gcd:  $\text{gcd} = 21$ .

(B.Tech note: can back-substitute to express gcd as linear combination  $21 = 105 - 42$  and  $42 = 252 - 2 \cdot 105$  if extended algorithm required.)

### (b) gcd (1220, 516)

1.  $1220 = 516 \cdot 2 + 188$ .
2.  $516 = 188 \cdot 2 + 140$ .
3.  $188 = 140 \cdot 1 + 48$ .
4.  $140 = 48 \cdot 2 + 44$ .
5.  $48 = 44 \cdot 1 + 4$ .
6.  $44 = 4 \cdot 11 + 0$ .

So  $\text{gcd} = 4$ .

You can back-substitute to find coefficients if asked (extended Euclid).

## 8. Induction proof: $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$

We prove by mathematical induction for  $n \geq 0$ .

- **Base step:**  $n = 0$ . LHS = 1. RHS =  $2^{0+1} - 1 = 2 - 1 = 1$ . Holds.
- **Inductive hypothesis:** suppose true for  $n = k$ :

$$1 + 2 + 2^2 + \dots + 2^k = 2^{k+1} - 1.$$

- **Inductive step:** add  $2^{k+1}$  to both sides:

$$1 + 2 + \dots + 2^k + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1} = 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1.$$

That proves the formula for  $n = k + 1$ .

By induction the identity holds for all  $n \geq 0$ .

(Alternative: derive as geometric series sum, but induction is requested.)

## 9. Number of license plates

Model: sequence of 3 uppercase letters followed by 3 digits. Letters: 26 choices each, digits: 10 each. Independent choices  $\Rightarrow$

$$\text{Total} = 26^3 \times 10^3 = (26 \cdot 26 \cdot 26) \times (10 \cdot 10 \cdot 10) = 17,576 \times 1,000 = 17,576,000.$$

## 10. Committee selection (10 men, 15 women; 6-member committee with equal numbers)

Equal men and women  $\Rightarrow$  3 men + 3 women.

Number of ways:

$$\binom{10}{3} \cdot \binom{15}{3}.$$

Compute:

$$\begin{aligned} \binom{10}{3} &= \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120. \\ \binom{15}{3} &= \frac{15 \cdot 14 \cdot 13}{6} = 455. \end{aligned}$$

Hence total  $120 \times 455 = 54,600$  committees.

(If order mattered, we would multiply by  $6!$ , but for committees order doesn't matter.)



## Assignment 2-1

### 1 — Poset questions (Figure-b, Hasse diagram)

**Note:** The PDF contains “Figure-b” but the diagram in the file is not legible in text form here. To avoid guessing the diagram, below I:

- give the **method** (how to obtain each requested item from any Hasse diagram), and
- show the **definitions / what to look for** (so you can apply them quickly to Figure-b), and if you upload the diagram image I will compute the exact answers.

#### Method / definitions (apply these to Figure-b):

- Maximal elements:** Elements with **no element strictly above them** (i.e., no upward edge from them in the Hasse diagram). Equivalently, an element  $x$  is maximal if there is no  $y \neq x$  with  $x \leq y$ .
- Minimal elements:** Elements with **no element strictly below them** (no downward edge to them in the diagram).  $x$  is minimal if there is no  $y \neq x$  with  $y \leq x$ .
- Greatest element (top):** An element  $g$  such that  $x \leq g$  for **every**  $x$  in the poset. In a Hasse diagram it is a single node above all others. If none exists, answer “no greatest element.”
- Least element (bottom):** An element  $l$  such that  $l \leq x$  for **every**  $x$ . In diagram it is a single node below all others.
- Upper bounds of  $\{a,b,c\}$ :** All elements  $u$  such that  $a \leq u$ ,  $b \leq u$ , and  $c \leq u$  (look for nodes reachable by upward chains from each of  $a,b,c$ ).
- Least upper bound (lub, join, supremum):** The smallest element among the upper bounds (an upper bound  $u$  is least if every other upper bound  $v$  satisfies  $u \leq v$ ). If multiple incomparable minimal upper bounds exist, the lub does not exist.
- Lower bounds of  $\{f,g,h\}$ :** All elements  $l$  such that  $l \leq f$ ,  $l \leq g$ , and  $l \leq h$  (look for nodes that are below all three).
- Greatest lower bound (glb, meet, infimum):** The largest element among these lower bounds (if exists; otherwise state none).

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If you'd like, upload a clear image or crop of **Figure-b** and I'll compute (a)–(h) explicitly. For now, proceed to the remaining questions (complete solutions below).

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### 2 — Four-digit numbers using digits $\{0,3,4,5,6,7\}$

We want 4-digit numbers (leading digit cannot be 0).

(i) **Repetition not allowed**

- Choices for first digit: any of  $\{3,4,5,6,7\} \rightarrow 5$  choices (0 disallowed).
- Remaining three places: choose without repetition from remaining 5 digits, then 4, then 3.

$$\text{Total} = 5 \times 5 \times 4 \times 3 = 300.$$

(ii) **Repetition allowed**

- First digit: 5 choices (3,4,5,6,7).
- Each of the other three positions: 6 choices (0,3,4,5,6,7) each.

$$\text{Total} = 5 \times 6 \times 6 \times 6 = 5 \cdot 6^3 = 5 \cdot 216 = 1080.$$

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### 3 — Converse, contrapositive, inverse

Proposition: “If 7 is greater than 5, then 8 is greater than 6.”

Let  $p = “7 > 5”$  and  $q = “8 > 6”$ . (Both  $p$  and  $q$  are **true** facts, but we list logical forms.)

- **Original (given):**  $p \rightarrow q$ : If  $7 > 5$  then  $8 > 6$ .
- **Converse:**  $q \rightarrow p$ : If  $8 > 6$  then  $7 > 5$ .
- **Contrapositive:**  $\neg q \rightarrow \neg p$ : If  $8 \leq 6$  then  $7 \leq 5$ .
- **Inverse:**  $\neg p \rightarrow \neg q$ : If  $7 \leq 5$  then  $8 \leq 6$ .

(Remember:  $p \rightarrow q$  is logically equivalent to its contrapositive  $\neg q \rightarrow \neg p$ , but not generally to its converse or inverse.)

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### 4 — Prove $\neg(p \vee q) \equiv \neg p \wedge \neg q$ by truth table (De Morgan)

Construct truth table for  $p, q$ :

| $p$ | $q$ | $p \vee q$ | $\neg(p \vee q)$ | $\neg p$ | $\neg q$ | $\neg p \wedge \neg q$ |
|-----|-----|------------|------------------|----------|----------|------------------------|
| T   | T   | T          | F                | F        | F        | F                      |
| T   | F   | T          | F                | F        | T        | F                      |
| F   | T   | T          | F                | T        | F        | F                      |
| F   | F   | F          | T                | T        | T        | T                      |

Compare columns  $\neg(p \vee q)$  and  $\neg p \wedge \neg q$ : they are identical for all 4 rows. Therefore  $\neg(p \vee q) \equiv \neg p \wedge \neg q$ .  $\square$

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### 5 — If $n$ is integer and $3n + 2$ is odd, then $n$ is odd

**Proof by contradiction (direct parity):**

Assume  $3n + 2$  is odd. Suppose, for contradiction, that  $n$  is even:  $n = 2k$  for some integer  $k$ . Then

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1),$$

which is even — contradiction. Therefore  $n$  cannot be even, so  $n$  must be odd.  $\square$

(Alternate direct proof: write odd number as  $2m + 1$  and solve for parity of  $n$ .)

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### 6 — Sum of two rational numbers is rational

Let  $r = \frac{a}{b}$  and  $s = \frac{c}{d}$  be rationals, where  $a, b, c, d$  are integers and  $b, d \neq 0$ . Then

$$r + s = \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}.$$

Since integers are closed under addition and multiplication,  $ad + bc$  and  $bd$  are integers and  $bd \neq 0$ . Hence  $r + s$  is a rational number.  $\square$

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### 7 — Use induction to prove the sum formula

The printed formula in the PDF looked garbled. The sequence shown is 10, 30, 50, .... The  $k$ -th term is  $10(2k - 1)$  (since terms are  $10 \cdot 1, 10 \cdot 3, 10 \cdot 5, \dots$ ). So the sum of the first  $n$  terms is

$$S_n = 10(1 + 3 + 5 + \dots + (2n - 1)).$$

But  $1 + 3 + \dots + (2n - 1) = n^2$ . Therefore

$$S_n = 10n^2.$$

I will prove  $S_n = 10n^2$  by induction.

**Base  $n = 1$ :**  $S_1 = 10$ . RHS  $10(1)^2 = 10$ . True.

**Inductive step:** Assume  $S_k = 10k^2$ . Then

$$S_{k+1} = S_k + \text{next term} = 10k^2 + 10(2(k+1) - 1) = 10k^2 + 10(2k+1).$$

Compute RHS:

$$10k^2 + 10(2k+1) = 10(k^2 + 2k + 1) = 10(k+1)^2.$$

Thus true for  $k+1$ . By induction  $S_n = 10n^2$  for all  $n \geq 1$ .

**Remark:** If your assignment intended the RHS  $\frac{n(2n-1)(2n+1)}{3}$  (as printed), that does **not** equal  $10n^2$  and does not match the left series 10,30,50,... — so I corrected to the consistent closed form  $10n^2$ . If the original problem meant something else, paste the exact expression and I'll adapt.

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**8 — Is  $(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$  a tautology, contradiction, or contingency?**

We check the formula:

$$\Phi = (p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r)).$$

**Informal logical proof that  $\Phi$  is always true (hence a tautology):**

Assume  $p \rightarrow (q \rightarrow r)$ . We must show  $(p \rightarrow q) \rightarrow (p \rightarrow r)$ . So assume  $p \rightarrow q$ . To prove  $p \rightarrow r$ , assume  $p$ . From  $p$  and  $p \rightarrow q$  we obtain  $q$ . From  $p$  and the hypothesis  $p \rightarrow (q \rightarrow r)$  we obtain  $q \rightarrow r$ . From  $q$  and  $q \rightarrow r$  we obtain  $r$ . Thus under the assumptions we reach  $r$ , so  $p \rightarrow r$  follows, and hence  $(p \rightarrow q) \rightarrow (p \rightarrow r)$  holds. Because this derivation does not require any special truth assignment,  $\Phi$  is true for all truth values of  $p, q, r$ . Therefore  $\Phi$  is a **tautology**.

(You can also confirm with a 8-row truth table; the informal derivation is succinct.)

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## 9 — Logical equivalences

**a) Show  $\neg p \leftrightarrow q$  and  $p \leftrightarrow \neg q$  are equivalent**

Observe:

$$\neg p \leftrightarrow q \text{ is logically the same as } q \leftrightarrow \neg p.$$

But equivalence is symmetric, and  $q \leftrightarrow \neg p$  is the same truth relation as  $p \leftrightarrow \neg q$  because swapping both sides and negating both sides preserves biconditional truth. Formally, both

statements assert “one of  $p, q$  is true exactly when the other is false”; they are simply different syntactic presentations of the same truth pattern.

**Truth-table check (compact):**

| $p$ | $q$ | $\neg p$ | $\neg q$ | $\neg p \leftrightarrow q$ | $p \leftrightarrow \neg q$ |
|-----|-----|----------|----------|----------------------------|----------------------------|
| T   | T   | F        | F        | F                          | F                          |
| T   | F   | F        | T        | T                          | T                          |
| F   | T   | T        | F        | T                          | T                          |
| F   | F   | T        | T        | F                          | F                          |

Columns  $\neg p \leftrightarrow q$  and  $p \leftrightarrow \neg q$  match row by row — hence equivalent.

**b) Show  $\neg(p \oplus q)$  and  $p \leftrightarrow q$  are equivalent**

Recall  $p \oplus q$  is XOR (true when exactly one of  $p, q$  is true). Its negation  $\neg(p \oplus q)$  is true exactly when  $p$  and  $q$  have the **same** truth value — which is exactly what  $p \leftrightarrow q$  (XNOR) states. Therefore  $\neg(p \oplus q) \equiv p \leftrightarrow q$ .

**Truth table (compact):**

| $p$ | $q$ | $p \oplus q$ | $\neg(p \oplus q)$ | $p \leftrightarrow q$ |
|-----|-----|--------------|--------------------|-----------------------|
| T   | T   | 0            | 1                  | 1                     |
| T   | F   | 1            | 0                  | 0                     |
| F   | T   | 1            | 0                  | 0                     |
| F   | F   | 0            | 1                  | 1                     |

Columns match  $\rightarrow$  equivalence proven.

## 10 — Truth tables for given compound propositions

I'll construct compact truth tables (all 4 combinations for 2 variables in (b); 16 rows for 4 variables for (a) but we can systematically fill them).

**(a)  $((p \rightarrow q) \rightarrow r) \rightarrow s$**

This depends on four variables  $p, q, r, s$ . I'll show the key intermediate columns and the final result — 16 rows total.

Let columns be:  $p, q, r, s, p \rightarrow q, (p \rightarrow q) \rightarrow r, ((p \rightarrow q) \rightarrow r) \rightarrow s$ .

I'll present them grouped (I'll list rows where  $(p \rightarrow q) \rightarrow r$  is computed and then final implication):

| p | q | r | s | $p \rightarrow q$ | $(p \rightarrow q) \rightarrow r$ | $((p \rightarrow q) \rightarrow r) \rightarrow s$ |
|---|---|---|---|-------------------|-----------------------------------|---------------------------------------------------|
| T | T | T | T | T                 | T                                 | T                                                 |
| T | T | T | F | T                 | T                                 | F                                                 |
| T | T | F | T | T                 | F                                 | T                                                 |
| T | T | F | F | T                 | F                                 | T                                                 |
| T | F | T | T | F                 | T                                 | T                                                 |
| T | F | T | F | F                 | T                                 | F                                                 |
| T | F | F | T | F                 | T                                 | T                                                 |
| T | F | F | F | F                 | T                                 | F                                                 |
| F | T | T | T | T                 | T                                 | T                                                 |
| F | T | T | F | T                 | T                                 | F                                                 |
| F | T | F | T | T                 | F                                 | T                                                 |
| F | T | F | F | T                 | F                                 | T                                                 |
| F | F | T | T | T                 | T                                 | T                                                 |
| F | F | T | F | T                 | T                                 | F                                                 |
| F | F | F | T | T                 | F                                 | T                                                 |
| F | F | F | F | T                 | F                                 | T                                                 |

(Explanation of sample rows: compute  $p \rightarrow q$ , then  $(p \rightarrow q) \rightarrow r$ , then final implication to s.)

You can use this table to read off the final value for every valuation.

### (b) $(p \wedge q) \rightarrow (p \vee q)$

This is simpler: two variables only.

| p | q | $p \wedge q$ | $p \vee q$ | $(p \wedge q) \rightarrow (p \vee q)$ |
|---|---|--------------|------------|---------------------------------------|
| T | T | T            | T          | T                                     |
| T | F | F            | T          | T                                     |
| F | T | F            | T          | T                                     |
| F | F | F            | F          | T                                     |

Every row yields **T**. So  $(p \wedge q) \rightarrow (p \vee q)$  is a **tautology** (intuitively because if both  $p$  and  $q$  are true then obviously at least one of them is true).